I find this view shortsighted, or, at the very least, premature. While we may never have technology capable of seeing strings directly, the history of science is replete with theories that were tested experimentally through indirect means.[13] String theory isn't modest. Its goals and promises are big. And that's exciting and useful, because if a theory is to be *the* theory of our universe, it must match the real world not just in the broad-brush outline discussed so far, but also in minute detail. As we'll now discuss, therein lie potential tests.

During the 1960s and 1970s, particle physicists made great strides in understanding the quantum structure of matter and the nongravitational forces that govern its behavior. The framework to which they were finally led by experimental results and theoretical insights is called the *standard model* of particle physics and is based on quantum mechanics, the matter particles in Table 12.1, and the force particles in Table 12.2 (ignoring the graviton, since the standard model does not incorporate gravity, and including the Higgs particle, which is not listed in the tables), all viewed as point particles. The standard model is able to explain essentially all data produced by the world's atom smashers, and over the years its inventors have been deservedly lauded with the highest of honors. Even so, the standard model has significant limitations. We've already discussed how it, and every other approach prior to string theory, failed to merge gravity and quantum mechanics. But there are other shortcomings as well.

The standard model failed to explain *why* the forces are transmitted by the precise list of particles in Table 12.2 and *why* matter is composed of the precise list of particles in Table 12.1. Why are there three families of matter particles and why does each family have the particles it does? Why not two families or just one? Why does the electron have three times the electric charge of the down-quark? Why does the muon weigh 23.4 times as much as the up-quark, and why does the top-quark weigh about 350,000 times as much as an electron? Why is the universe constructed with this range of seemingly random numbers? The standard model takes the particles in Tables 12.1 and 12.2 (again, ignoring the graviton) as *input*, then makes impressively accurate predictions for how the particles will interact and influence each other. But the standard model can't explain the input—the particles and their properties—any